

BART Application

This file: we already had optimal lag solved in Kaggle, based on that `opt_p`, and all other parameter assumed, here target to fit in specific models and forecast for our variables locally

```
R_packages <- c("dbarts", "tidyverse", "lubridate", "knitr", "tibble")

options(repos = c(CRAN = "https://cloud.r-project.org"))

if (!requireNamespace("librarian", quietly = TRUE))
  install.packages("librarian")

librarian::shelf(R_packages)
set.seed(12345)
```

1. DATA as usual

```
df_ml <- read.csv("balanced_can_md.csv")
df_ml$GDP_per_w <- df_ml$GDP_new - df_ml$EMP_CAN
df_ml %>%
  select(Date, GDP_new, EMP_CAN, GDP_per_w) %>%
  head(10)
```

##	Date	GDP_new	EMP_CAN	GDP_per_w
## 1	1981-01-01	0.001274909	0.005951733	-0.004676824
## 2	1981-02-01	0.003042663	0.006829789	-0.003787126
## 3	1981-03-01	-0.000819862	-0.000407765	-0.000412097
## 4	1981-04-01	0.011764570	0.003283988	0.008480582
## 5	1981-05-01	-0.008363640	0.002647652	-0.011011292
## 6	1981-06-01	-0.001484003	0.002623079	-0.004107082
## 7	1981-07-01	-0.003620080	-0.003451923	-0.000168157
## 8	1981-08-01	-0.008109328	0.001533702	-0.009643030
## 9	1981-09-01	0.011234802	-0.002213161	0.013447963
## 10	1981-10-01	-0.007063600	-0.002527754	-0.004535846

```
targets <- c("GDP_per_w", "CPI_ALL_CAN", "NHOUSE_P_CAN")

features_large <- c(
  "GDP_per_w", "CPI_ALL_CAN", "NHOUSE_P_CAN", "IP_new",
  "BSI_new", "UNEMP_CAN", "EMP_CONS_CAN", "hstart_CAN_new",
  "CRED_HOUS", "CRED_HOUS_MORT", "BANK_RATE_L", "TBILL_3M",
  "Exp_BP_new", "Imp_BP_new", "IPPI_METAL_CAN", "OILP_new",
  "WTISPLC"
)
```

2. Helper

```
build_lagged_panel <- function(df, predictors, target, p, h) {
  dat <- df %>%
    dplyr::select(Date, all_of(unique(c(predictors, target)))) %>%
    arrange(Date)

  for (lag in 1:p) {
    for (v in predictors) {
      dat[[paste0("L", lag, "_", v)]] <- dplyr::lag(dat[[v]], lag)
    }
  }

  dat[[paste0("F", h, "_", target)]] <- dplyr::lead(dat[[target]], h)

  dat_final <- dat %>% drop_na()

  X <- as.matrix(dat_final %>% select(starts_with("L")))
  Y <- as.matrix(dat_final %>% select(starts_with("F")))

  list(X = X, Y = Y)
}
```

```
target <- "CPI_ALL_CAN"
predictors <- features_large
p_short <- 11
p_long <- 3
```

```
ntrees_direct <- 500
nskip_direct <- 1000
ndposts_direct <- 2000
```

```
bart_direct_h <- function(df, predictors, target, p, h) {

  panel <- build_lagged_panel(df, predictors, target, p, h)
  X <- panel$X
  y <- panel$Y[, 1]

  fit <- dbarts::bart(
    x.train = X,
    y.train = y,
    ntree = ntrees_direct,
    nskip = nskip_direct,
    ndpost = ndposts_direct,
    keeptrees = TRUE,
    verbose = FALSE
  )

  X_last <- tail(X, 1)
  pred <- predict(fit, X_last)

  as.numeric(mean(pred))
}
```

```

}

horizons_short <- 1:3

bart_short_fc_direct <- sapply(horizons_short, function(h) {
  cat("Direct BART short, h =", h, "\n")
  bart_direct_h(df_ml, predictors, target, p_short, h)
})

## Direct BART short, h = 1
## Direct BART short, h = 2
## Direct BART short, h = 3

horizons_long <- 1:12

bart_long_fc_direct <- sapply(horizons_long, function(h) {
  cat("Direct BART long, h =", h, "\n")
  bart_direct_h(df_ml, predictors, target, p_long, h)
})

## Direct BART long, h = 1
## Direct BART long, h = 2
## Direct BART long, h = 3
## Direct BART long, h = 4
## Direct BART long, h = 5
## Direct BART long, h = 6
## Direct BART long, h = 7
## Direct BART long, h = 8
## Direct BART long, h = 9
## Direct BART long, h = 10
## Direct BART long, h = 11
## Direct BART long, h = 12

last_date <- as.Date(tail(df_ml$Date, 1))

bart_short_tbl <- tibble(
  h      = horizons_short,
  date   = seq(last_date %m+% months(1), by = "1 month", length.out = 3),
  forecast = bart_short_fc_direct
)

bart_long_tbl <- tibble(
  h      = horizons_long,
  date   = seq(last_date %m+% months(1), by = "1 month", length.out = 12),
  forecast = bart_long_fc_direct
)

knitr::kable(bart_short_tbl, digits = 4,
             caption = "Direct BART CPI forecasts (h = 1-3)")

```

Table 1: Direct BART CPI forecasts (h = 1–3)

h	date	forecast
1	2025-08-01	0.0018
2	2025-09-01	0.0020
3	2025-10-01	0.0018

```
knitr::kable(bart_long_tbl, digits = 4,
             caption = "Direct BART CPI forecasts (h = 1-12)")
```

Table 2: Direct BART CPI forecasts (h = 1–12)

h	date	forecast
1	2025-08-01	0.0013
2	2025-09-01	0.0016
3	2025-10-01	0.0018
4	2025-11-01	0.0023
5	2025-12-01	0.0027
6	2026-01-01	0.0021
7	2026-02-01	0.0008
8	2026-03-01	-0.0009
9	2026-04-01	0.0017
10	2026-05-01	0.0020
11	2026-06-01	0.0030
12	2026-07-01	0.0014

```
targets_to_run <- c("GDP_per_w", "CPI_ALL_CAN", "NHOUSE_P_CAN")

all_short <- list()
all_long <- list()

for (tt in targets_to_run) {
  fc_s <- sapply(1:3, function(h) {
    bart_direct_h(df_ml, predictors, tt, p_short, h)
  })
  all_short[[tt]] <- fc_s

  fc_l <- sapply(1:12, function(h) {
    bart_direct_h(df_ml, predictors, tt, p_long, h)
  })
  all_long[[tt]] <- fc_l
}

last_date <- as.Date(tail(df_ml$Date, 1))

bart_short_all <- tibble(
  variable = rep(targets_to_run, each = 3),
  h         = rep(1:3, times = length(targets_to_run)),
  date      = rep(seq(last_date %m+% months(1), by = "1 month", length.out = 3),
    times = length(targets_to_run)),
  forecast = unlist(all_short)
```

```

)

bart_long_all <- tibble(
  variable = rep(targets_to_run, each = 12),
  h        = rep(1:12, times = length(targets_to_run)),
  date     = rep(seq(last_date %m+% months(1), by = "1 month", length.out = 12),
    times = length(targets_to_run)),
  forecast = unlist(all_long)
)

knitr::kable(bart_short_all, digits = 4,
  caption = "Direct BART Forecasts (ALL VARIABLES, h = 1-3)")

```

Table 3: Direct BART Forecasts (ALL VARIABLES, h = 1–3)

variable	h	date	forecast
GDP_per_w	1	2025-08-01	0.0008
GDP_per_w	2	2025-09-01	0.0007
GDP_per_w	3	2025-10-01	0.0015
CPI_ALL_CAN	1	2025-08-01	0.0017
CPI_ALL_CAN	2	2025-09-01	0.0019
CPI_ALL_CAN	3	2025-10-01	0.0017
NHOUSE_P_CAN	1	2025-08-01	-0.0007
NHOUSE_P_CAN	2	2025-09-01	-0.0007
NHOUSE_P_CAN	3	2025-10-01	-0.0002

```

knitr::kable(bart_long_all, digits = 4,
  caption = "Direct BART Forecasts (ALL VARIABLES, h = 1-12)")

```

Table 4: Direct BART Forecasts (ALL VARIABLES, h = 1–12)

variable	h	date	forecast
GDP_per_w	1	2025-08-01	0.0012
GDP_per_w	2	2025-09-01	0.0014
GDP_per_w	3	2025-10-01	0.0016
GDP_per_w	4	2025-11-01	-0.0003
GDP_per_w	5	2025-12-01	0.0007
GDP_per_w	6	2026-01-01	0.0015
GDP_per_w	7	2026-02-01	0.0021
GDP_per_w	8	2026-03-01	0.0014
GDP_per_w	9	2026-04-01	0.0002
GDP_per_w	10	2026-05-01	0.0002
GDP_per_w	11	2026-06-01	0.0007
GDP_per_w	12	2026-07-01	0.0016
CPI_ALL_CAN	1	2025-08-01	0.0012
CPI_ALL_CAN	2	2025-09-01	0.0015
CPI_ALL_CAN	3	2025-10-01	0.0020
CPI_ALL_CAN	4	2025-11-01	0.0022
CPI_ALL_CAN	5	2025-12-01	0.0027
CPI_ALL_CAN	6	2026-01-01	0.0020

variable	h	date	forecast
CPI_ALL_CAN	7	2026-02-01	0.0008
CPI_ALL_CAN	8	2026-03-01	-0.0006
CPI_ALL_CAN	9	2026-04-01	0.0016
CPI_ALL_CAN	10	2026-05-01	0.0021
CPI_ALL_CAN	11	2026-06-01	0.0028
CPI_ALL_CAN	12	2026-07-01	0.0014
NHOUSE_P_CAN	1	2025-08-01	-0.0007
NHOUSE_P_CAN	2	2025-09-01	-0.0013
NHOUSE_P_CAN	3	2025-10-01	-0.0009
NHOUSE_P_CAN	4	2025-11-01	-0.0003
NHOUSE_P_CAN	5	2025-12-01	0.0002
NHOUSE_P_CAN	6	2026-01-01	-0.0003
NHOUSE_P_CAN	7	2026-02-01	0.0008
NHOUSE_P_CAN	8	2026-03-01	0.0004
NHOUSE_P_CAN	9	2026-04-01	0.0015
NHOUSE_P_CAN	10	2026-05-01	0.0009
NHOUSE_P_CAN	11	2026-06-01	0.0016
NHOUSE_P_CAN	12	2026-07-01	0.0009

```
write.csv(bart_short_all, "BART_all_variables_short.csv", row.names = FALSE)
write.csv(bart_long_all, "BART_all_variables_long.csv", row.names = FALSE)
```

ignore right now, old code not delete yet

```
## hard code gen nice table lol
comparison <- data.frame(
  Model      = c("Small VAR (3 vars)",
                 "Large VAR (17 vars)",
                 "Random Forest",
                 "BART"),
  RMSE_1to3  = c(0.003892, 0.0030008, 0.002372, 0.00266),
  RMSE_1to12 = c(0.003624, 0.0034235, 0.002554, 0.002717)
)

knitr::kable(comparison, digits=6,
             caption = "Model Out-of-Sample-CV RMSE: VAR vs RF vs BART")
```

Table 5: Model Out-of-Sample-CV RMSE: VAR vs RF vs BART

Model	RMSE_1to3	RMSE_1to12
Small VAR (3 vars)	0.003892	0.003624
Large VAR (17 vars)	0.003001	0.003424
Random Forest	0.002372	0.002554
BART	0.002660	0.002717

```
write.csv(
  bart_short_tbl,
  "BART_forecasts_short.csv",
  row.names = FALSE
```

```
)  
  
write.csv(  
  bart_long_tbl,  
  "BART_forecasts_long.csv",  
  row.names = FALSE  
)  
  
write.csv(  
  comparison,  
  "Model_Comparison_RMSE.csv",  
  row.names = FALSE  
)
```